Lattice-Based Blind Signatures for Privacy Tokens in E-Business Systems: A Preliminary Design and Analysis*

Hao-En Hou and Chun-I Fan†
Department of Computer Science and Engineering, National Sun Yat-sen University, Taiwan
pqc143040082@gmail.com, cifan@mail.cse.nsysu.edu.tw

Abstract

Digital commerce platforms increasingly rely on coupons, loyalty credits, membership benefits, and access tokens. However, in many current deployments, the issuance and redemption of tokens remain operationally linkable, enabling platforms or merchants to correlate user activity across transactions. Blind signatures offer a natural way to authorize a token without revealing its final redeemable form to the signer, while recent progress in lattice-based blind signatures suggests that this functionality can be studied in a post-quantum setting. This poster investigates how lattice-based blind signatures can serve as the foundation of privacy tokens for e-business systems. We focus on anonymous coupon and loyalty-token redemption, define a simple issuer-user-verifier model, identify the core requirements of blindness, unlinkability, unforgeability, double spending prevention, and deployability, and explain why this application setting exposes practical trade-offs that are not fully captured by primitive-level efficiency results alone. Compared with existing privacy-token frameworks such as Privacy Pass, lattice-based blind signatures may still impose larger artifacts and communication costs, making deployment evaluation especially important.

* Proceedings of The 2026 IFIP WG 8.4 International Symposium on E-Business Information Systems Evolution (EBISION 2026), Article No. P-31, July 8-10, 2026, Seoul, South Korea.
© The copyright of this paper remains with the author(s).
† Corresponding author

1 Motivation and Research Gap

Blind signatures were originally introduced for untraceable payments and remain relevant to modern anonymous-token systems [1]. Similar privacy concerns arise in digital commerce whenever a platform issues redeemable objects and later observes or influences how they are used. Promotional coupons, loyalty claims, membership rewards, limited-use access passes, and event-specific benefits all illustrate this pattern. In conventional deployments, the same operator or a tightly coupled group of providers may see both issuance and redemption events, making user activity linkable across time. This is problematic when a business needs token validity and anti-abuse controls, but not full behavioral traceability. Recent work on lattice-based blind signatures has mainly focused on construction quality, security, round complexity, and efficiency [2], [3], while token-oriented systems such as Privacy Pass have shown that anonymous issuance and redemption are practical architectural concerns [4], [5]. What remains underexplored is how lattice-based blind signatures should be positioned within a concrete e-business privacy-token workflow and how deployment-relevant factors such as artifact size and communication overhead should be evaluated in that setting.

2 Privacy-Token Model and Design Perspective

We consider a three-party setting with an issuer or platform, a user, and a verifier such as a merchant or service endpoint. During issuance, the user requests a token associated with a redeemable business benefit. The request is blinded before reaching the issuer, allowing the issuer to authorize issuance without learning the final token instance that will later be presented. After receiving the blind signature, the user unblinds it and stores a valid token. During redemption, the user presents the token to the verifier, which checks validity and enforces usage constraints such as single-use or replay detection. The goal is not absolute anonymity against every party under all assumptions; rather, it is practical unlinkability between issuance and redemption while preserving business-side controls against forgery and repeated use. Our analysis therefore centers on five requirements: blindness, unlinkability, unforgeability, double spending prevention, and deployability. Lattice-based blind signatures are appealing because they point toward post-quantum resilience while preserving authorization without direct token disclosure [2], [3]. At the same time, larger signatures, communication cost, protocol complexity, and integration overhead may complicate adoption in real e-business infrastructures, especially when recent constructions can still reach artifacts in the tens-of-kilobytes range.

3 Expected Contribution

This work argues that lattice-based blind signatures should be studied not only as standalone cryptographic primitives, but also as candidate building blocks for application-layer privacy tokens in future digital commerce. The poster makes three preliminary contributions. First, it defines a concrete e-business scenario centered on the anonymous redemption of coupons and loyalty tokens. Second, it maps the main properties of lattice-based blind signatures to application-level requirements for privacy, fraud prevention, and operational feasibility. Third, it complements existing privacy-token frameworks by highlighting the additional deployment considerations introduced by post-quantum lattice-based designs, including artifact size, communication overhead, and system-level feasibility. The intended outcome is not a finalized construction or deployment recipe, but a clearer research agenda for quantum-safe privacy tokens and a more precise understanding of the trade-offs that must be addressed to make this direction practical.

References

[1] D. Chaum, “Blind Signatures for Untraceable Payments”, in *Advances in Cryptology, Springer, 1983.*
[2] S. Agrawal, E. Kirshanova, D. Stehlé, and A. Yadav, “Practical, Round-Optimal Lattice-Based Blind Signatures”, in *CCS ’22, 2022*.
[3] W. Beullens, V. Lyubashevsky, N. K. Nguyen, and G. Seiler, “Lattice-Based Blind Signatures: Short, Efficient, and Round-Optimal”, in *CCS ’23, 2023*.
[4] A. Davidson, J. Iyengar, and C. A. Wood, “The Privacy Pass Architecture”, RFC 9576, 2024.
[5] S. Celi, A. Davidson, S. Valdez, and C. A. Wood, “Privacy Pass Issuance Protocols”, RFC 9578, 2024.